Calibrated Ensemble Estimation of Weekly Wildfire Occurrence Risk Along Power-Grid Corridors from Open Data

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Abstract

Vegetation in contact with electrical infrastructure is a leading cause of both catastrophic wildfires and large-scale outages, yet operational tools that prioritize *which corridors to inspect this week* remain rare, and the data behind them is typically proprietary. We present Firefinder, a deployed national-scale system that estimates weekly wildfire occurrence probability for every $\sim$5 km^2 hexagonal cell of Portugal, Spain and France, and aggregates that risk onto 304,758 power line corridors extracted from OpenStreetMap. The system is built exclusively from free, keyless, public data: raw Sentinel-2 imagery with in-pipeline cloud masking, ERA5-derived weather, a decade of Copernicus EFFIS burnt-area perimeters as labels, and open terrain and land-cover products. Wildfire occurrence is an extreme rare event in this framing, with weekly per-cell base rates between 1.5×10^{-3} and 8.1×10^{-5}; we therefore evaluate exclusively on strict temporal holdouts using ranking metrics, and convert scores to usable likelihoods with isotonic calibration. A seed-diverse gradient-boosted ensemble improves holdout PR-AUC in Portugal (+29%) and France (+61%) while degrading it in Spain, a region-dependence we report rather than average away; ROC-AUC across the three countries ranges from 0.84 to 0.91. A pairwise ranking objective wins on ROC-AUC in two regions but never on top-of-list precision, and we discuss why calibrated classification remains preferable for corridor triage. Member disagreement provides a per-cell uncertainty signal at no additional cost. The full system, including the daily refresh pipeline and the web application, is open source at `https://github.com/davidcoallier/firefinder.eu` and deployed at `https://firefinder.eu`.

1 Introduction

Climate change is lengthening fire seasons and intensifying fire weather across southern Europe [1]. At the same time, electrical grids are both a frequent ignition source and a primary casualty of wildfire: vegetation contact with conductors ignites fires, and fires in turn force outages and preventive shutoffs. Utilities and civil-protection agencies consequently need *corridor-level* prioritization, knowing which spans of the network deserve attention this week, rather than regional fire-danger indices that describe broad areas.

Machine learning approaches, particularly gradient-boosted trees such as XGBoost [2], have shown strong performance in wildfire prediction [3–5], but published systems typically stop at gridded danger maps, rely on curated or proprietary inputs, and report metrics that obscure the extreme class imbalance of the task. Recent work has also highlighted that rare-event fire prediction degrades under the distribution shift between fire seasons [6], an effect we observe directly in our per-year results.

This paper describes Firefinder, an end-to-end open system with four contributions:

1. **A national-scale, fully open pipeline.** Every input is free, public and keyless: raw Sentinel-2 L2A scenes (processed with our own cloud masking into monthly vegetation composites), ERA5-derived daily weather with an automatic fallback provider, EFFIS burnt-area perimeters, Copernicus DEM terrain, ESA WorldCover fuels, and the OpenStreetMap power network. The pipeline covers 226,839 cells across three countries and refreshes scores daily on free public CI infrastructure.

2. **Honest rare-event methodology.** We formalize the task as long-tail binary prediction on cell-weeks, evaluate only on strict temporal holdouts (train through 2023, test 2024–2026), report PR-AUC against base rates of 10^{-3} to 10^{-5}, and calibrate scores to true likelihoods with isotonic regression, showing that even the highest-risk cells rarely exceed a few percent weekly occurrence probability.

3. **Corridor-level risk aggregation.** Cell risk is mapped onto the power network with an explicit, documented aggregation rule, producing a ranked list of corridors with per-corridor explanations derived from ensemble-averaged SHAP attributions [7].

4. **An ensemble ablation with a negative result.** Seed-diverse ensembling improves PR-AUC substantially; a pairwise ranking objective improves ROC-AUC but *harms* PR-AUC, and we argue why it is the wrong trade for triage products despite ranking being the product surface.

2 Problem Formulation

Let $\mathcal{G}$ be the set of H3 resolution-7 hexagonal cells [8] whose centers fall within a country polygon, and let t index ISO weeks of the April–October fire season. For each cell-week (g, t) we construct a feature vector $x_{g,t} \in \mathbb{R}^{21}$ (Section 3) and a binary label

$$y_{g,t} = \mathbf{1}\left[\exists\, \text{EFFIS perimeter } F : \text{date}(F) \in t \ \wedge \ F \cap g \neq \emptyset\right],$$

that is, whether an official burnt-area perimeter with a fire date in week t intersects cell g. The task is to estimate $p_{g,t} = \Pr(y_{g,t} = 1 \mid x_{g,t})$.

Two aspects of this target deserve precision. First, $y_{g,t}$ is *wildfire occurrence*, not ignition: a single large fire produces many positive cells from one ignition point, so per-cell probabilities describe the chance a cell is touched by fire that week, which for corridor triage is arguably the more operationally relevant event. We use “occurrence” throughout and reserve “ignition” for ignition-source discussions. Second, the weekly weather features are *observed*: retrospective results measure discrimination given the week’s realized weather, and live operation substitutes the most recent observations (persistence). We therefore speak of risk estimation rather than forecasting; substituting numerical weather prediction inputs would make the system a true forecast without changing the learning problem.

The label distribution is extremely long-tailed. Test-period base rates $\bar{y}$ are 1.5×10^{-3} (Portugal), 3.8×10^{-4} (Spain) and 8.1×10^{-5} (France): between 1-in-700 and 1-in-12,000. Accuracy is therefore uninformative, and random splits leak: adjacent weeks in the same cell are nearly identical, so shuffled cross-validation grossly overstates skill. All evaluation uses temporal holdout, training on seasons through 2023 and testing on 2024–2026, which additionally exposes the model to genuine seasonal distribution shift [6, 9].

The base learner minimizes class-weighted empirical risk

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\left[w_+\, y_i \log \hat{p}_i + (1 - y_i)\log(1 - \hat{p}_i)\right], \qquad w_+ = \frac{N_-}{N_+}, \tag{1}$$

Table 1: Dataset statistics. Cell-weeks are April–October rows with a usable vegetation composite; positives are cell-weeks intersecting an EFFIS perimeter. Corridors are OSM power line segments of at most 5 km.

Region	Cells	Corridors	Cell-weeks	Positives	Test base rate
Portugal	16,342	57,864	3.10M	3,357	1.50×10^{-3}
Spain	95,497	60,159	16.53M	4,492	3.79×10^{-4}
France	115,000	186,735	19.10M	1,105	8.1×10^{-5}

with w_+ the inverse class ratio, implemented as `scale_pos_weight` in XGBoost.

3 Data and Features

Analysis grid. Cells are H3 resolution 7 (~5.16 km^2), clipped to Natural Earth country polygons rather than bounding boxes. All raster inputs are warped onto a common ~200 m grid per region, then aggregated per cell. Table 1 summarizes the regions.

Vegetation state (4 features). From raw Sentinel-2 L2A scenes on AWS Open Data we build monthly composites: scenes are cloud- and shadow-masked using the scene classification layer (keeping only vegetated and bare classes), and per-pixel means are accumulated in a streaming pass so country-scale months fit in memory. Per cell we take mean NDVI, the 10th percentile of NDVI (driest patch), mean NDMI (moisture), and an NDVI anomaly against the cell's own calendar-month climatology. Critically, week t uses the latest composite *strictly before* t: using the same month would leak the post-fire NDVI collapse into the features.

Fire weather (8 features). Daily ERA5-derived variables from Open-Meteo (NASA POWER as automatic fallback) on a 0.25–0.5° grid, aggregated to the week: maximum temperature, minimum relative humidity, maximum sustained wind and gusts, precipitation sum, reference evapotranspiration, and trailing 30- and 90-day precipitation ending the day before the week as drought proxies. Weekly weather is *observed*, a stand-in for a forecast product; backtests are unaffected but live scores assume persistence (Section 6).

Static surface (7 features). Elevation and slope from the Copernicus GLO-30 DEM; tree, shrub, grass and crop cover fractions from ESA WorldCover as fuel proxies; and distance from the cell center to the nearest power line.

Seasonality (2 features). Sine and cosine encodings of ISO week.

4 Method

4.1 Seed-diverse gradient-boosted ensemble

We train $K = 5$ XGBoost classifiers $f_1, \ldots, f_K$ that share hyperparameters (400 trees, depth 6, learning rate 0.05, column subsample 0.8) but differ in random seed and row subsample ratio (0.7–0.9). The ensemble score is the mean probability

$$\bar{p}(x) = \frac{1}{K}\sum_{k=1}^{K} f_k(x), \qquad s(x) = \Big(\tfrac{1}{K}\sum_{k=1}^{K}\big(f_k(x) - \bar{p}(x)\big)^2\Big)^{1/2}, \tag{2}$$

where the member standard deviation $s(x)$ is retained as a per-cell disagreement signal and surfaced in the application, distinguishing confident risk from contested risk at no additional training cost.

4.2 Isotonic calibration

Class weighting (Eq. 1) deliberately distorts probabilities to improve minority-class learning, so raw scores are rankings, not likelihoods. We fit an isotonic regression [10] $g^* = \arg\min_{g\in\mathcal{M}} \sum_i \big(g(\bar{p}_i) - y_i\big)^2$ over the class of monotone step functions $\mathcal{M}$, fitted *forward*: only the first held-out season (2024) is used to fit g^*, and reliability is reported on the untouched 2025–2026 seasons. Because g^* is monotone, all ranking metrics are unchanged; what changes is interpretability. After calibration, the single highest-risk cell-week in Portugal carries roughly a 5% weekly occurrence probability, an honest number that reframes the product interface: tiers are assigned by position in the week's risk distribution, while calibrated percentages are shown as likelihoods.

4.3 Corridor aggregation

The OSM power network is segmented into corridors of at most 5 km. For corridor c, let $C(c)$ be the cells whose centers lie within 500 m of the line. The corridor score is

$$R(c) = 0.65 \max_{g\in C(c)} \bar{p}_g \;+\; 0.35\,\frac{1}{|C(c)|}\sum_{g\in C(c)} \bar{p}_g. \tag{3}$$

The weights are a documented design heuristic, not fitted parameters: fire exposure along a corridor does not average (one severe crossing is actionable), while the mean term breaks ties toward sustained exposure. Fitting these weights would require corridor-level incident labels, which do not exist publicly in Europe. Corridors are ranked nationally per week, and each carries the explanation of its riskiest cell, computed as the mean SHAP attribution across ensemble members so explanations match the deployed score.

4.4 Pairwise ranking objective (experiment)

Since the product surface is a ranking, we also train a `rank:pairwise` XGBoost ranker with weeks as query groups, an analogue of preference-based objectives recently proposed for wildfire prediction [6] in the gradient-boosted setting.

5 Experiments

Setup. All models train on fire seasons through 2023 and are evaluated on 2024–2026 (partial 2026). The single/ensemble/ranker ablation is run identically in all three regions. We report PR-AUC (average precision) and ROC-AUC; PR-AUC should be read against each region's base rate, and we report the ratio as lift.

Table 2: Held-out performance of the deployed K=5 calibrated ensembles (2024–2026). Lift is PR-AUC divided by the test base rate.

Region	PR-AUC	ROC-AUC	Base rate	Lift
Portugal	0.0174	0.844	1.50×10^{-3}	11.6×
Spain	0.0108	0.861	3.79×10^{-4}	28.5×
France	0.0029	0.906	8.2×10^{-5}	35.4×

Table 3: Ablation on identical temporal holdouts, all regions. Ensembling improves PR-AUC in Portugal and France but not Spain; the pairwise ranker never wins on PR-AUC.

Region	Model	PR-AUC	ROC-AUC
Portugal	Single XGBoost	0.0135	0.842
	Ensemble (K=5)	**0.0174**	0.844
	Pairwise ranker	0.0088	**0.850**
Spain	Single XGBoost	**0.0153**	0.856
	Ensemble (K=5)	0.0108	**0.861**
	Pairwise ranker	0.0051	0.822
France	Single XGBoost	0.0018	0.904
	Ensemble (K=5)	**0.0029**	0.906
	Pairwise ranker	0.0011	**0.910**

Main results. Table 2 summarizes held-out performance. ROC-AUC between 0.84 and 0.90 across three distinct national fire regimes indicates the feature set generalizes; PR-AUC lifts of 12–26× over base rate quantify practical triage value at extreme imbalance. Figure 1 shows precision-recall behavior for the Portugal ablation.

Ensembling helps where it matters. The K=5 ensemble improves PR-AUC by 29% over its strongest single member (Table 3) with negligible ROC-AUC change: variance reduction concentrates its benefit at the top of the score distribution, exactly the region a triage product consumes. Median member disagreement in a scored week is comparable to the score scale itself, underlining why a single tree ensemble's point estimate should not be over-read.

A ranking objective is the wrong trade. The pairwise ranker takes the best ROC-AUC in Portugal and France yet posts the worst PR-AUC in every region, and in Spain loses on both metrics. Pairwise losses optimize ordering across the entire distribution, spending capacity separating mid-tail negatives; the classifier ensemble concentrates on the extreme tail. Since corridor triage consumes only the top of the list, we retain the calibrated classifier ensemble in production and report the ranker as a cautionary result: optimizing the metric that resembles the product surface is not the same as optimizing the product.

Calibration. Figure 1 (right) shows the reliability diagram under the forward design: the calibrator is fitted on 2024 and evaluated on untouched 2025–2026. Raw class-weighted scores overstate probability by roughly an order of magnitude; the forward-fitted isotonic transform aligns predicted probability with observed frequency on unseen seasons across the probability range. Post-calibration, the distribution of deployed probabilities is honest but compressed (maximum $\approx$ 0.05 in a typical Portuguese week), motivating the distribution-relative tier presentation described in Section 4.

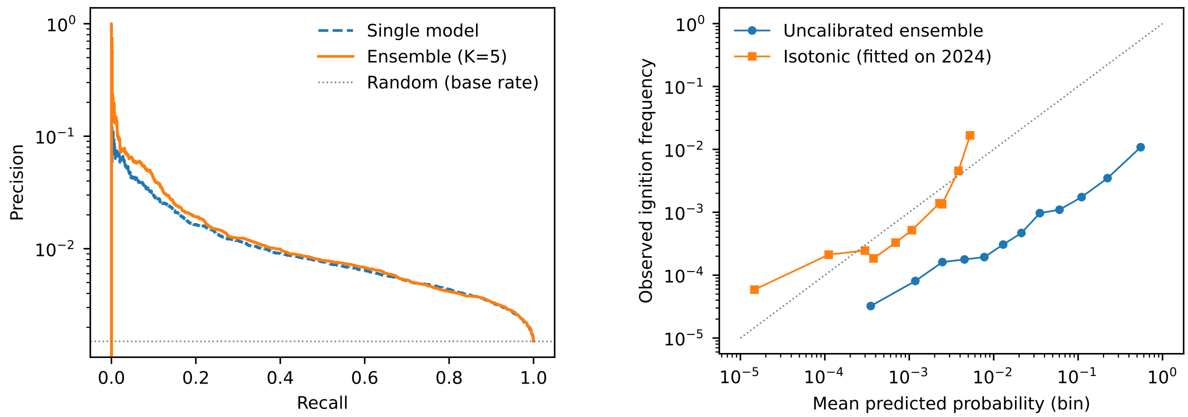

Figure 1: Portugal temporal holdout (2024–2026). Left: precision-recall curves (log precision axis); the ensemble dominates the single model in the high-precision regime. Right: reliability diagram on quantile bins (log-log), with the isotonic calibrator fitted on the 2024 season and evaluated on untouched 2025–2026; class weighting inflates raw scores by roughly an order of magnitude, and the forward-fitted transform aligns them with observed frequencies on unseen seasons.

Seasonal shift dominates. Per-year results (Figure 2, right) vary far more than model-choice deltas: Portugal reaches ROC-AUC 0.889 and PR-AUC 0.0395 in the severe 2025 season but 0.80 in the quieter 2024 and (partial) 2026 seasons, a direct observation of the distribution-shift problem highlighted by Jiang and Sun [6]. Models are most reliable exactly when fire load is high, which is operationally fortunate, but closing the quiet-season gap, for example through season-similarity weighted ensembling, is the clearest future-work direction.

6 Limitations

Weekly weather features are observed rather than forecast; a production forecast system would substitute NWP outputs, leaving backtests unchanged. EFFIS perimeters are MODIS-derived and under-report small fires, so labels are a conservative subset of true fires, and label noise differs by country. The evaluation is temporally but not spatially blocked; spatial autocorrelation within a week may flatter absolute metrics, though it affects all compared models equally. The corridor aggregation weights in Eq. 3 are heuristic pending corridor-level incident data, which no public European dataset provides: corridor risk is fire risk near the line, not a claim that the line causes ignition. Finally, the label is burnt-area occurrence rather than point ignition, so spatially large fires contribute many positive cells from a single ignition; models of ignition points proper would require ignition-location datasets that EFFIS does not provide.

7 Conclusion

Firefinder demonstrates that operationally useful, corridor-level wildfire risk estimation can be built end to end from free public data and modest models, provided the rare-event statistics are treated honestly: temporal holdouts, base-rate-aware metrics, calibration to true likelihoods, and uncertainty from ensemble disagreement. A seed-diverse XGBoost ensemble with isotonic calibration currently offers the best precision-at-the-top of the alternatives tested, and a pairwise ranking objective, despite matching the prod-

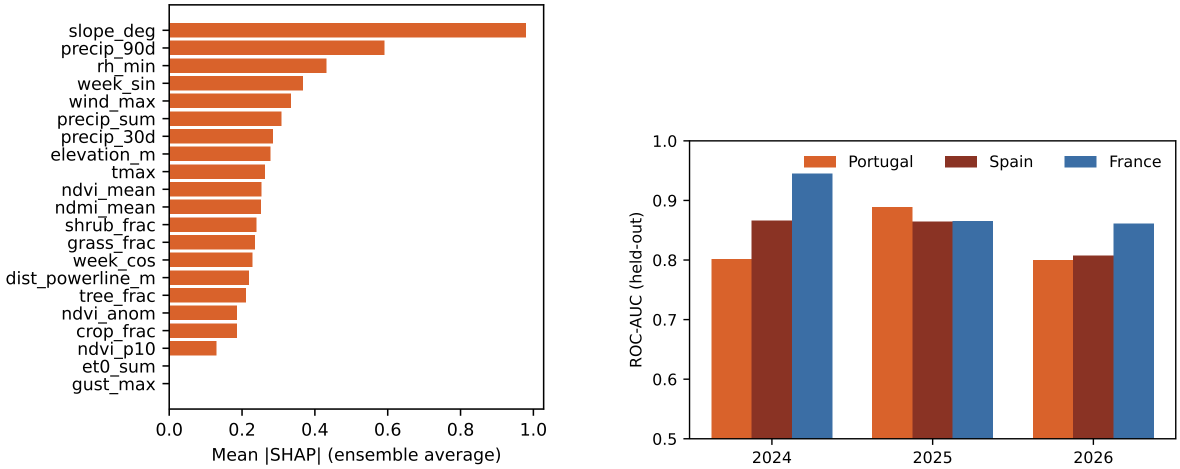

Figure 2: Left: ensemble-averaged mean |SHAP| feature importances on the Portugal holdout. Right: held-out ROC-AUC by test year and region; season-to-season variance is the dominant error source.

uct surface, trades away exactly the precision the product needs. The system, data pipeline and models are open source, and the deployed application rescores three countries daily at `https://firefinder.eu`.

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